Formality of $\mathbb{E}_n$ and knot theory

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Plan:

1. Spaces of long knots

2. A crash course on mfld-calculus

3. A few words about Bousfield-Kan spect. seq. (BK-SS)

4. Multiplicative operads and Sinha's cosimplicial space

5. Formality implies collapse of BK-SS at E^2

6. Vassiliev invariants of knots

1. Spaces of long knots

Let us fix $d \geq 3$

(Def) The space of long knots in $\mathbb{R}^d$ is

$$\mathrm{Emb}_c(\mathbb{R},\mathbb{R}^d) := \left\{ K : \mathbb{R} \hookrightarrow \mathbb{R}^d \text{ smooth embedding} \;:\; K = \epsilon \text{ outside } \underbrace{[-1,1]}_{\square^1} \right\}$$

ϵ : unknot : $\mathbb{R} \hookrightarrow \mathbb{R}^d$, $t \mapsto (t,0\dots0)$

equipped with weak $\mathcal{C}^\infty$-topology

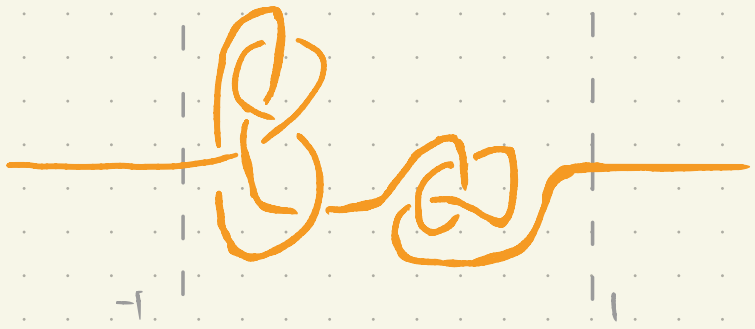
Rks:
- $\mathrm{Emb}_c(\mathbb{R},\mathbb{R}^d) \simeq \mathrm{Emb}_\partial(\square^1,\square^d)$
- $\pi_0 \mathrm{Emb}_c(\mathbb{R},\mathbb{R}^3) \cong \pi_0 \mathrm{Emb}(S^1,S^3)$ ~ isot. classes of "classical" knots

(1 pt-compact.)

Goal: Study invariants of this space, such as: $H_*^{\mathbb{Q}}$, $\pi_*^{\mathbb{Q}}$

$H_*^{\mathbb{Q}} = H_*(-,\mathbb{Q})$ $\quad$ $\pi_*^{\mathbb{Q}} = \pi_*(-)\otimes\mathbb{Q}$

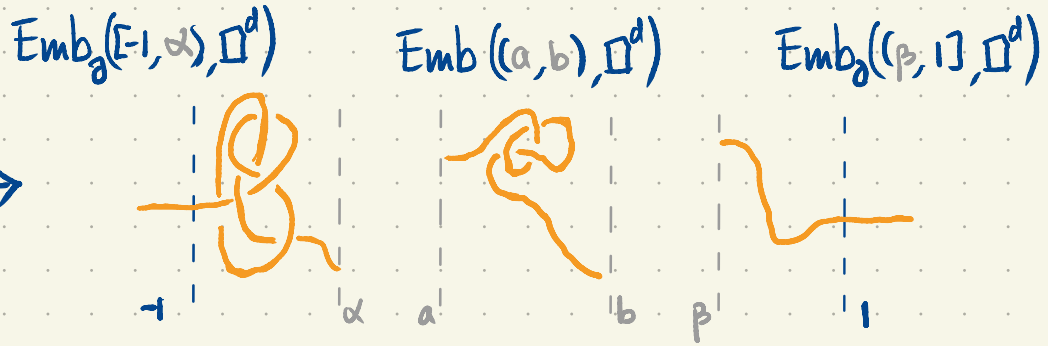
Picture: $K \in \mathrm{Emb}_c(\mathbb{R}, \mathbb{R}^d)$ looks like



even $(\mathrm{mfld}^{\partial,\mathrm{fr}}_{1/\square^1})^{\mathrm{op}} \to \mathrm{Spc}$

We can upgrade this space to a functor $\mathrm{Emb}_c(-, \mathbb{R}^d) : \mathrm{Open}^{\partial}(\square^1)^{\mathrm{op}} \to \mathrm{Spc}$ (opens $\supseteq \partial\square^1$)

$\mathrm{Emb}_c(-, \mathbb{R}^d)$: $[-1, \alpha) \sqcup (a, b) \sqcup (\beta, 1] \mapsto$



Upshot: we can exploit this functoriality to split $\mathrm{Emb}_c(\mathbb{R}, \mathbb{R}^d)$ up in terms of our old friends: the little n-discs operads $\mathbb{E}_n$

2. Manifold, Goodwillie-Weiss or GW calculus

This successful theory was designed to study "good" functors

$$\mathcal{F}: \mathrm{Mfld}_m^{op} \longrightarrow \mathrm{Spc} \quad (\text{resp. } \mathcal{F}: \mathrm{Open}(M)^{op} \longrightarrow \mathrm{Spc})$$

via successive multi-local approximations; i.e. restrictions for $k \geq 0$

$$\mathcal{F}_{|k}: (\mathrm{Discs}_m^{\leq k})^{op} \longrightarrow \mathrm{Spc} \quad (\text{resp. } \mathcal{F}_{|1}: \mathrm{Discs}^{\leq k}(M)^{op} \longrightarrow \mathrm{Spc})$$

$\rightsquigarrow$ opens $= \coprod_{i \in I} D_i$ s.t. $\#I \leq k$

The idea is that extending $\mathcal{F}_{|k}$ back to Mfld_m, we get the degree $\leq k$ universal approximation of $\mathcal{F}$:

$$\mathcal{F}(M) \longrightarrow T_k\mathcal{F}(M) := \underset{\mathrm{Discs}_m^{\leq k}/M}{\mathrm{holim}} \mathcal{F}_{|k}$$

Σ

deg $\leq k$ univ. appr. evaluated at $M \in \mathrm{Mfld}_m$

Rks: • deg$\leq k$ approx. $\sim$ deg$\leq k$ Taylor polynomial

$\mathcal{F} \longrightarrow T_k\mathcal{F}$ $\qquad$ $f \rightsquigarrow P_k(f)$ $\qquad$ Idea: deg$\leq k$ polynomial is completely

$(\mathcal{F}: \mathrm{Mfld}_m^{op} \to \mathcal{S}pc)$ $\qquad$ $(f: \mathbb{R} \to \mathbb{R})$ $\qquad$ characterised by value on $(k+1)$-pts

• These "approximations" can be interpreted as sheafifications for certain

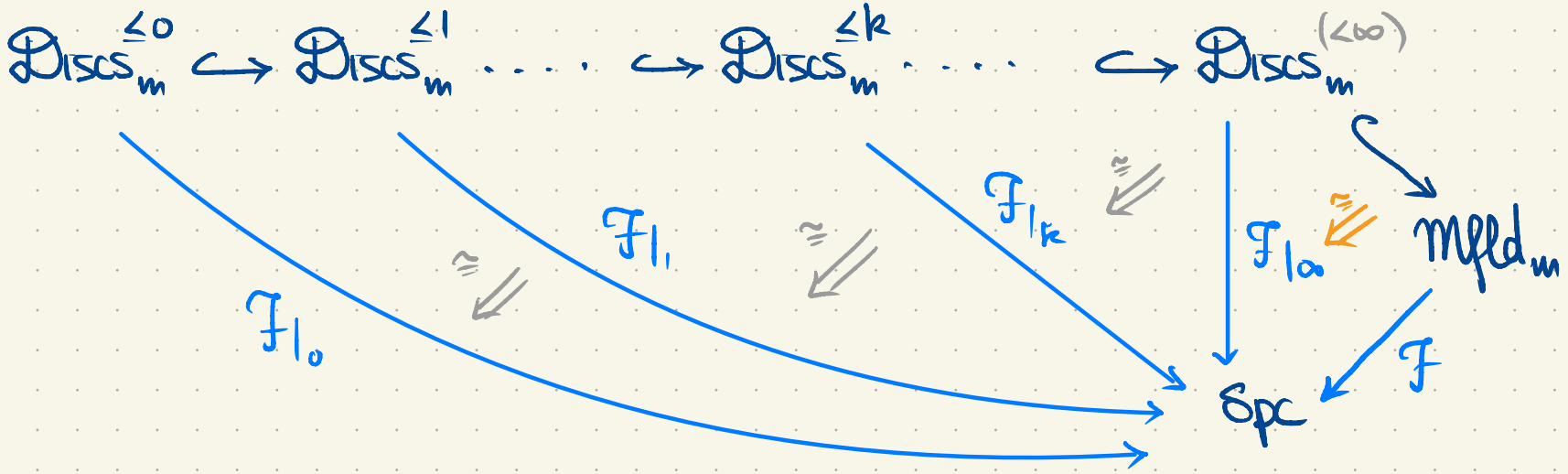
$\mathcal{J}_k$-topologies: $(U_i)_{i\in I}$ $\mathcal{J}_k$-cover of M if $\forall S \subseteq M$ w/ $\#S \leq k$, $S \subseteq U_i$ for some i

• $T_k\mathcal{F}$ is derived Ran along $\mathrm{Discs}_m^{\leq k} \overset{\iota_k}{\hookrightarrow} \mathrm{Mfld}_m$. Because of this, there is a derived mapping formula for $T_k\mathcal{F}$:

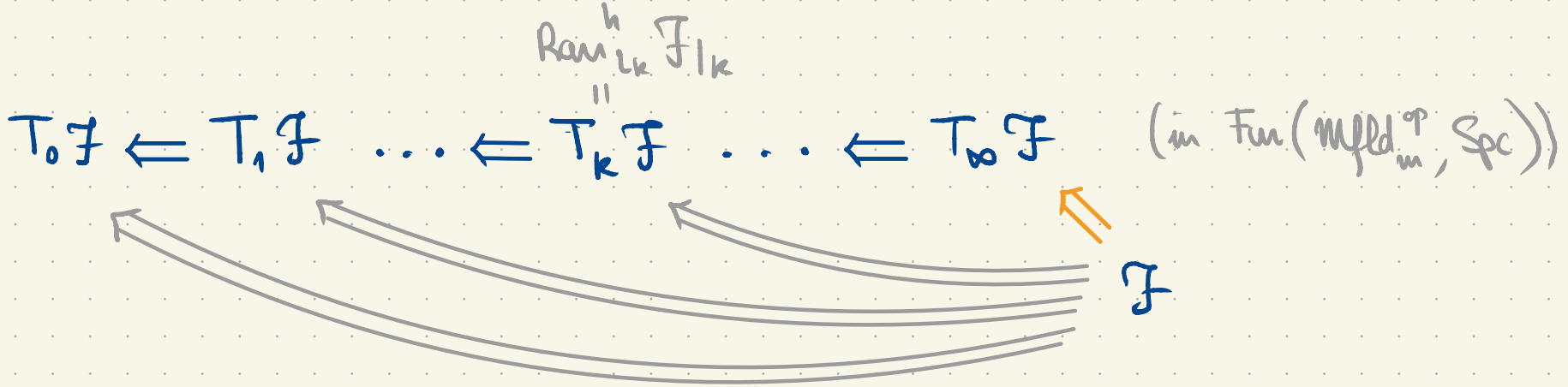
$$T_k\mathcal{F}(M) := \mathrm{Ran}^h_{\iota_k}\mathcal{F}|_k \simeq \mathrm{Map}^h_{\mathrm{Discs}_m^{\leq k}}\left(\mathrm{emb}_M|\,,\ \mathcal{F}|\right)$$

$\mathrm{emb}_M : V \mapsto \mathrm{Emb}(V, M)$

Given a (good) functor $\mathcal{F} : \mathrm{Mfld}_m^{op} \to \mathcal{S}pc$, we can form a diagram



Extending everything to Mfld_m we get a tower (GW-tower) of approx.



Main idea: get to know $\mathcal{F}$ using the GW-tower:

$$T_0\mathcal{F} \leftarrow T_1\mathcal{F} \cdots \leftarrow T_k\mathcal{F} \cdots \leftarrow T_\infty\mathcal{F}$$

Best case scenario: $\mathcal{F} \xrightarrow{\simeq} T_\infty\mathcal{F}$ natural weak homotopy equivalence

(or even $\mathcal{F} \xrightarrow{\simeq} T_k\mathcal{F}$ for $k<\infty$)

Thm (Goodwillie-Klein, Boavida-Weiss) Let N be a smooth mfld (w/o bdry).

Then,

$$emb_N \xrightarrow{\simeq} T_\infty emb_N \quad \text{in } Fun(Mfld_m^{op}, Spc) \text{ for } \dim N - m \geq 3$$

$$\| \\ Map^h_{Discs_m}(emb_{\mathbb{R}^m}, emb_N)$$

In other words, $Emb(M,N) \xrightarrow{\simeq} Map^h_{Discs_m}(emb_M, emb_N)$ if $codim_N M \geq 3$.

⚠ In our case, $\mathrm{Emb}_c(\mathbb{R},\mathbb{R}^d)$ does not directly fit into this framework.

(the problem is the "cpt-support" condition)

Instead, $\mathrm{Emb}_c(-,\mathbb{R}^d)\colon \mathrm{Open}^{\partial}(\square^1)^{op} \longrightarrow \mathrm{Spc}$ (or even $(\mathrm{Mfld}^{\partial,fr}_{1/\square^1})^{op} \longrightarrow \mathrm{Spc}$)

opens $\supseteq \partial\square^1$

Luckily, there is a parallel story suitable for this example:

$$T_0\,\mathrm{Emb}_c(-,\mathbb{R}^d) \Leftarrow \cdots \Leftarrow T_k\,\mathrm{Emb}_c(-,\mathbb{R}^d) \Leftarrow \cdots \Leftarrow T_\infty\,\mathrm{Emb}_c(-,\mathbb{R}^d)$$

(in $\mathrm{Fun}((\mathrm{Mfld}^{\partial,fr}_{1/\square^1})^{op}, \mathrm{Spc})$)

$\mathrm{Emb}_c(-,\mathbb{R}^d)$

Actually, we are interested in the value at $\square^1$, so we just give it a name

$$T_0\,\mathcal{K}nots_d \longleftarrow \cdots \longleftarrow T_k\,\mathcal{K}nots_d \longleftarrow \cdots \longleftarrow T_\infty\,\mathcal{K}nots_d$$

(in Spc)

$\mathrm{Emb}_c(\mathbb{R},\mathbb{R}^d) =: \mathcal{K}nots_d$

Thm (Goodwillie-Klein) For $d \geq 4$ (i.e. $\operatorname{codim}_{\mathbb{R}^d} \mathbb{R} \geq 3$)

$$\mathcal{K}nots_d \xrightarrow{\sim} T_\infty \mathcal{K}nots_d \quad \text{is a weak homotopy equivalence}$$

⚠ The bound $d \geq 4$ is sharp. However, we can still say something in $d = 3$.

(Vassiliev invariants of knots)

Q: How is this useful?

A: We know how to use towers of spaces to compute invariants (spectral seqs)

Main goal: Obtain information about $H_*^{\mathbb{Q}}(\mathcal{K}nots_d)$ & $\pi_*^{\mathbb{Q}}(\mathcal{K}nots_d)$

from relative $\mathbb{Q}$-(co)formality of $\mathbb{E}_d$

Hint: For $k \geq 2$,

$$\begin{array}{ccc} T_k \mathcal{K}nots_d & \longrightarrow & \Omega \mathrm{Lin}^{\mathrm{mono}}(\mathbb{R}, \mathbb{R}^d) \simeq \Omega \mathrm{Map}^h_{\mathrm{Opd}_{\leq 2}}(\mathbb{E}_1, \mathbb{E}_d) \\ \downarrow & \lrcorner^h & \downarrow \\ * & \longrightarrow & \Omega \mathrm{Map}^h_{\mathrm{Opd}_{\leq k}}(\mathbb{E}_1, \mathbb{E}_d) \end{array}$$

3. Bousfield-Kan spectral sequences (BK-SS)

So, we have a nice tower in Spc

$$T_0\,\mathcal{K}nots_d \longleftarrow \cdots \longleftarrow T_k\,\mathcal{K}nots_d \longleftarrow \cdots \longleftarrow T_\infty\,\mathcal{K}nots_d$$

$\simeq$ (for $d \geq 4$)

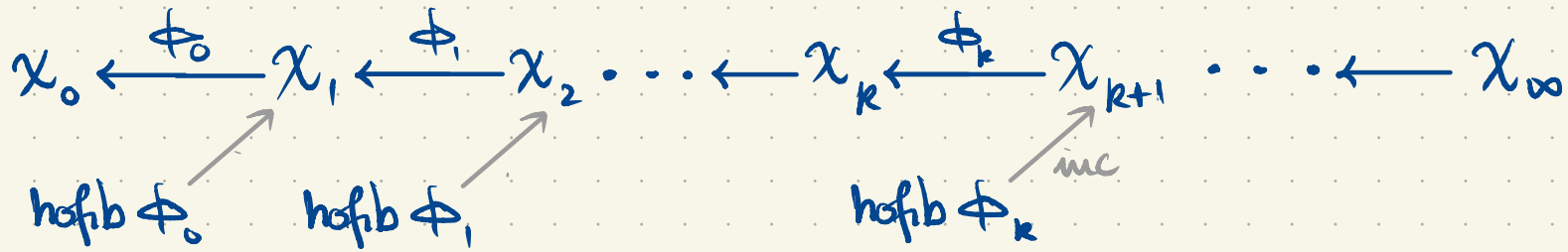
$\mathcal{K}nots_d$

Let us now review how this can be used to compute invt's of $\mathcal{K}nots_d$

Let's take a step back and simply start from a tower of Spc_*

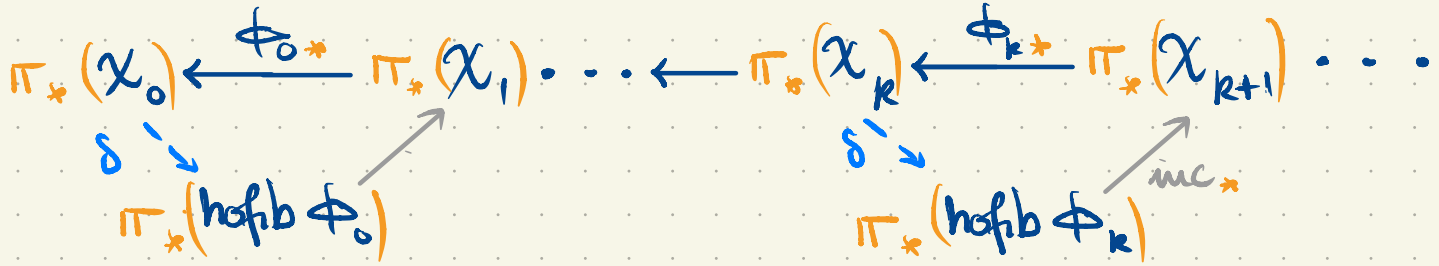
$$X_0 \xleftarrow{\phi_0} X_1 \xleftarrow{\phi_1} X_2 \cdots \longleftarrow X_k \xleftarrow{\phi_k} X_{k+1} \cdots \longleftarrow X_\infty \simeq \operatorname{holim}_k X_k$$

Take homotopy fibers at all the stages:



Then we obtain interconnected fiber sequences which yield

interconnected l.e.s. of invt's (just take your favorite; e.g. π_*)

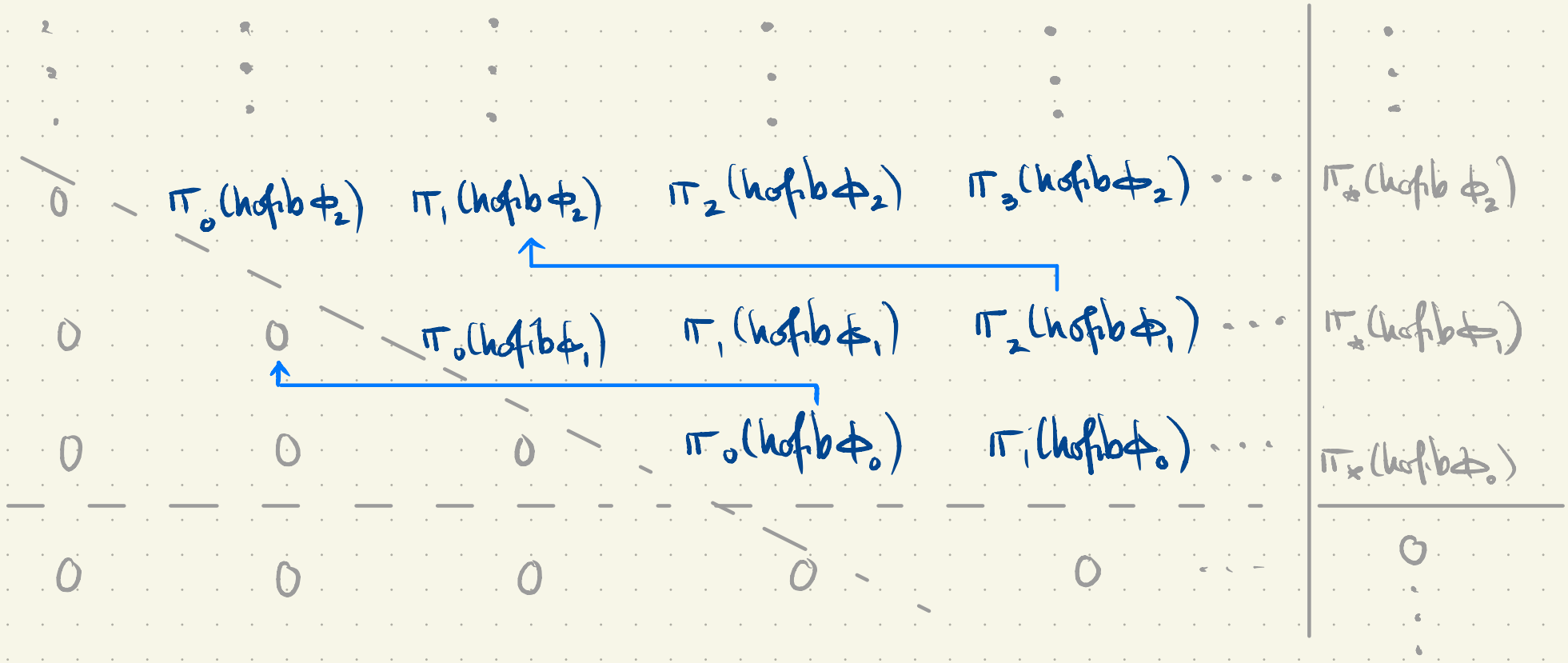


This is the data of an exact couple; we have found the π_*-BK spec.seq.

$$E^1_{-s,t} := \pi_{t-s}(\mathrm{hofib}\,\phi_s) \Longrightarrow \lim_k \pi_{t-s}(X_k)$$

cond. conv.

Picture: E_1-page of the spectral sequence



The differential is:

$$d_1 : \pi_i(\mathrm{hofib}\,\phi_k) \xrightarrow{\mathrm{inc}_*} \pi_i(X_k) \xrightarrow{\delta} \pi_{i-1}(\mathrm{hofib}\,\phi_{k+1})$$

A couple of technical remarks:

(1) All the invariants we will consider are ab. gps (not always automatically)

(2) We will always have strong convergence for the SS' (whatever that means)

(3) The SS will converge to the invariants we are interested in:

$E^1 \underset{\text{st. conv.}}{\Longrightarrow} \varprojlim_k H_{t-s}(X_k)$, but this guy is related to $H_{t-s}(\operatorname{holim}_k X_k)$:

Milnor s.e.s.: $0 \to \varprojlim_k{}^1 H_{t-s-1}(X_k) \to H_{t-s}(\operatorname{holim}_k X_k) \to \varprojlim_k H_{t-s}(X_k) \to 0$

0 in our examples $\Rightarrow H_{t-s}(X_\infty) \cong \varprojlim_k H_{t-s}(X_k)$

Moral: we are in a very nice scenario =) Don't worry too much

Meta-thm: In the cases of interest to us,

$$\pi_*\text{-BKSS} \overset{\text{str. conv.}}{\rightsquigarrow} \pi_*(X_\infty) \quad \& \quad H_*\text{-BKSS} \overset{\text{str. conv.}}{\rightsquigarrow} H_*(X_\infty)$$

(The rational versions as well, of course, $\pi_*^{\mathbb{Q}}$ & $H_*^{\mathbb{Q}}$)

A notion that will appear several times in the sequel:

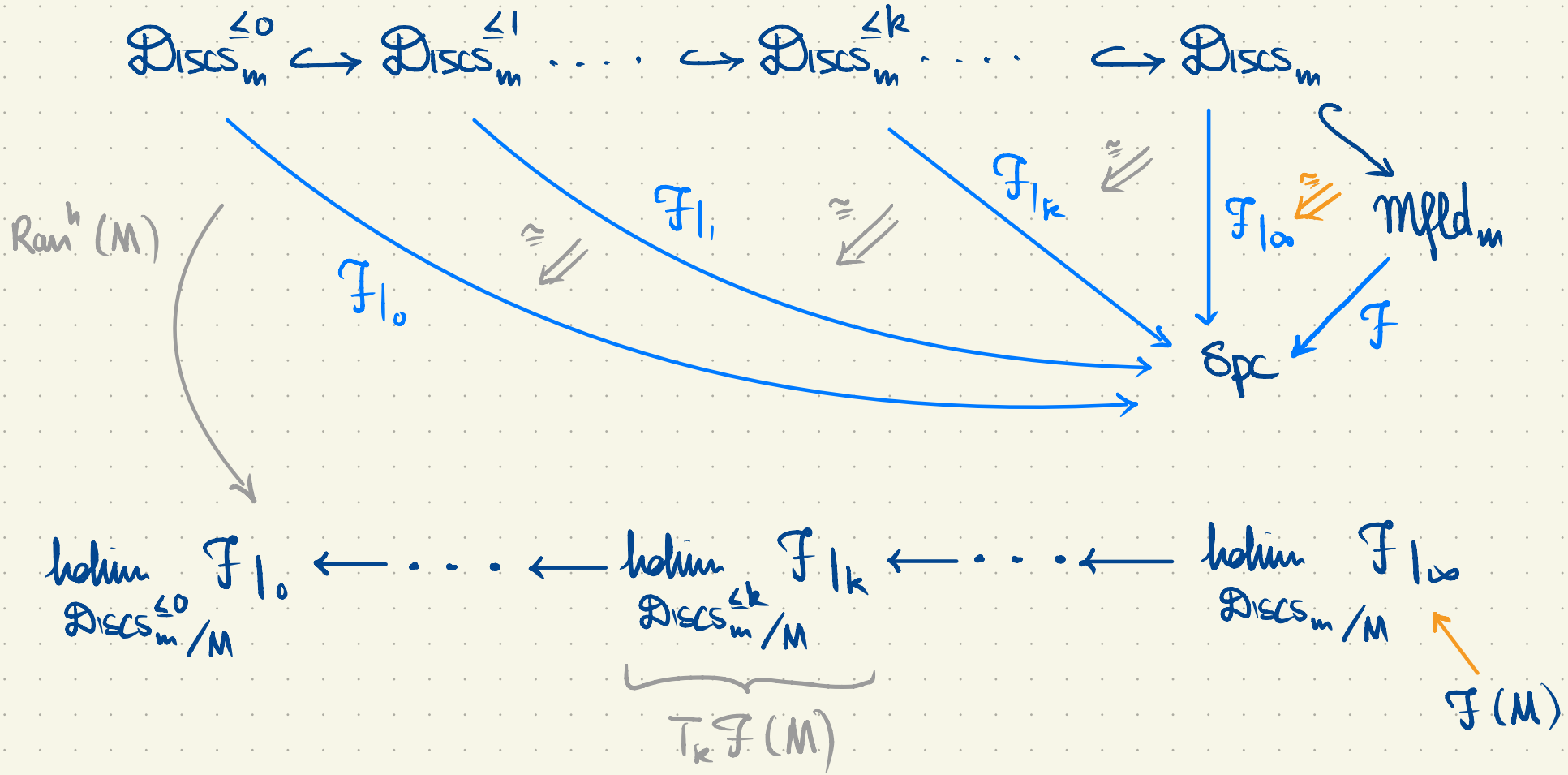
(Def) A spectral sequence $\{E^r, d_r\}_{r\geq 1}$ collapses at E^s if

$$E^s_{*,*} \cong E^{s+1}_{*,*} \cong \cdots \cong E^\infty_{*,*} \qquad (\text{e.g. } d_t = 0 \ \forall t \geq s)$$

$$\| \\ H(E^s, d_s)$$

Before moving forward, let us mention another source of towers of spaces

Recall: mfld-calculus produces towers of spaces : $\mathcal{F} \in \mathrm{Fun}(\mathrm{Mfld}_m^{op}, \mathrm{Spc})$



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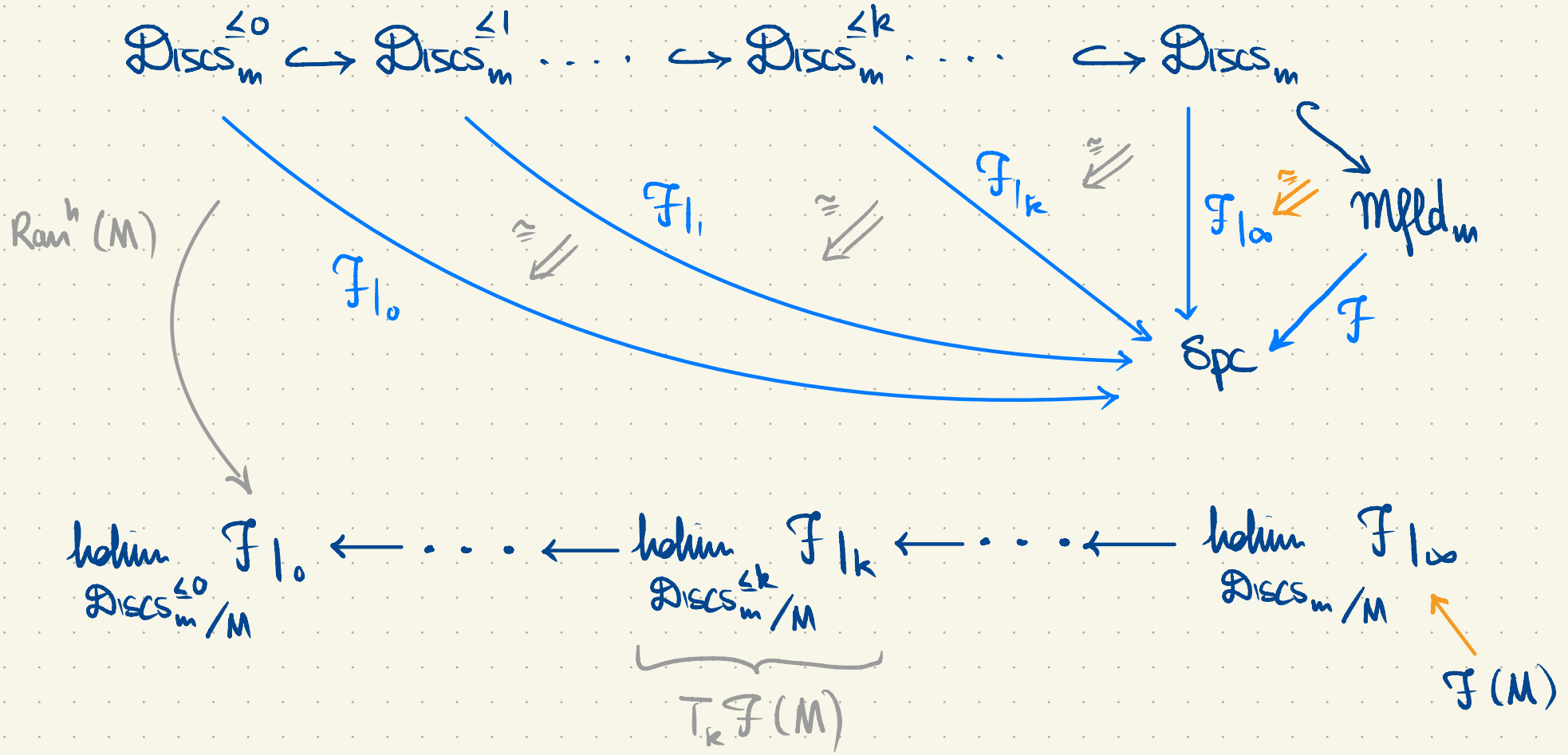

Now we consider a similar thing, but starting from a cosimplicial spc

$K^\bullet \in \mathrm{Spc}^\Delta$

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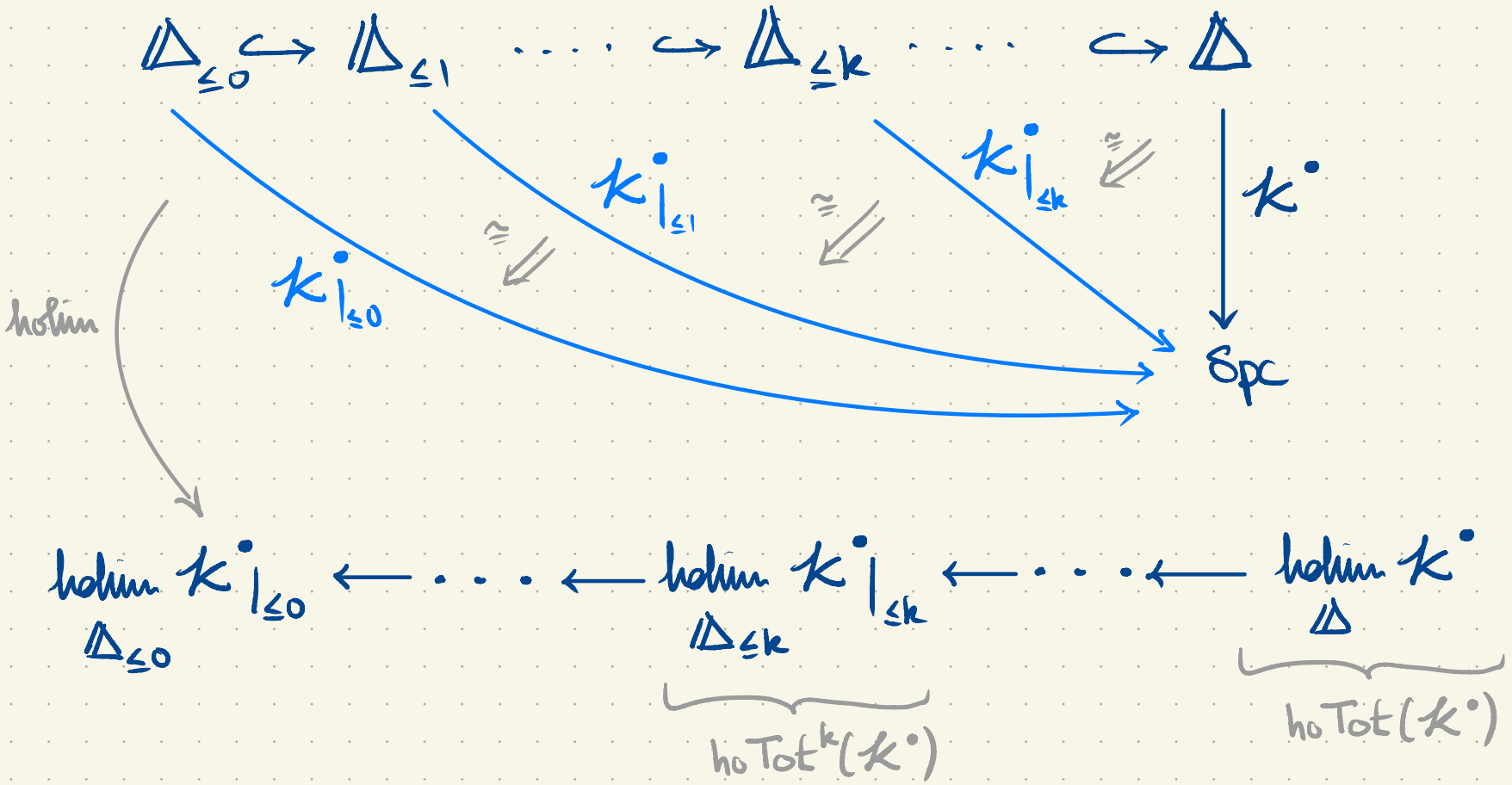

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$K^\bullet \in \mathrm{Spc}^{\Delta}$

For towers associated to (nice) cosimplicial spaces, there is a simple criterion to check if $H^{\mathbb{Q}}_*/\pi^{\mathbb{Q}}_*$- BK-spectral sequence collapses.

(Def) $\mathcal{K}^\bullet : \Delta \to Spc$ is $\mathbb{Q}$-formal if $A_{PL}(\mathcal{K}^\bullet) \simeq \cdots \simeq H^*(A_{PL}(\mathcal{K}^\bullet))$

($cdga_{\mathbb{Q}}$-model) nat.-qis' $\simeq$ $H^*_{\mathbb{Q}}(\mathcal{K}^\bullet)$

Prop. If $\mathcal{K}^\bullet : \Delta \to Spc$ is nice enough,

$\mathcal{K}^\bullet$ $\mathbb{Q}$-formal $\Rightarrow$ $H^{\mathbb{Q}}_*$-BK spect.seq. collapses at E^2

(Proof) By def. $C^{\mathbb{Q}}_*(\mathcal{K}^\bullet) \simeq A_{PL}(\mathcal{K}^\bullet)^\vee : \Delta \to Ch(\mathbb{Q})_{\geq 0}$ is formal

$$\Rightarrow \quad \underbrace{\left\{E^r_{BK}(A_{PL}(\mathcal{K}^\bullet)^\vee)\right\}_{r\geq 1}}_{\text{BK-spectral sequence}} \cong \underbrace{\left\{E^r_{BK}(H^*(A_{PL}(\mathcal{K}^\bullet))^\vee)\right\}_{r\geq 1}}_{\text{this ss collapses at } E^2 \text{ (no vertical differen.)}}$$

Now the Koszul dual statement:

(Def) $\mathcal{K}^\bullet : \Delta \to \mathrm{Spc}_*^{\geq 2}$ is $\mathbb{Q}$-coformal if $\lambda(\mathcal{K}^\bullet) \simeq \cdots \simeq H_*(\lambda(\mathcal{K}^\bullet))$

($\mathrm{dgl}_{\mathbb{Q}}$-model) (nat. qis.) (SII $\pi_*^{\mathbb{Q}}(\Omega\mathcal{K}^\bullet)$)

Prop. If $\mathcal{K}^\bullet : \Delta \to \mathrm{Spc}_*^{\geq 2}$ is even nicer, ([Arone et al.] assume f.d. conditions)

$\mathcal{K}^\bullet$ $\mathbb{Q}$-coformal $\Rightarrow$ $\pi_*^{\mathbb{Q}}$-BK spect.seq. collapses at E^2

Rk: We are safe, our examples satisfy the technical conditions above

([ALTV] define $\mathbb{Q}$-coformality as $A_{PL}(\mathcal{K}^\bullet) \simeq CE^*(\underbrace{\mathrm{Prim}\, H_*^{\mathbb{Q}}(\Omega\mathcal{K}^\bullet)}_{\pi_*^{\mathbb{Q}}(\Omega\mathcal{K}^\bullet)})$

4. Back to long-knots: Sinha's cosimplicial space $\mathcal{K}^\bullet_d$

Goal: Show that $H^{\mathbb{Q}}_\star/\pi^{\mathbb{Q}}_\star$-BK spect.seq. associated to

$$T_0\,\mathcal{K}nots_d \longleftarrow \cdots \longleftarrow T_k\,\mathcal{K}nots_d \longleftarrow \cdots \longleftarrow T_\infty\,\mathcal{K}nots_d \xleftarrow{\ \sim\ } \mathcal{K}nots_d \qquad d\geq 4$$

collapses at E^2

Idea: Relate this tower to one assoc. to a $\mathbb{Q}$-(co)formal cosimplicial space

⚠ It is possible to stick w/ $\mathcal{K}nots_d$ as defined so far, but we will consider instead: $\overline{\mathcal{K}nots}_d \equiv \overline{Emb_c}(\mathbb{R},\mathbb{R}^d) := Emb_c(\mathbb{R},\mathbb{R}^d)\times\Omega^2 S^{d-1}$

This change is harmless and makes the discussion closer to $\overline{E}_d$

The source of this cosimpl. space is actually operadic and super general:

(Def) A multiplicative $\mathcal{V}$-operad is a pair $(\mathcal{O}, As_+ \xrightarrow{\eta} \mathcal{O}^{ns})$:

- $\mathcal{O} \in Opd(\mathcal{V})$ (sym. operad in $\mathcal{V}$)
- $As_+ \in nsOpd(\mathcal{V})$ of unital assoc. algs.
- $\eta : As_+ \longrightarrow \mathcal{O}^{ns}$ map in $nsOpd(\mathcal{V})$

($\mathcal{O}^{ns}$: forget Σ-action)

Since As_+ is generated by $m =$ & $e =$ this is the same as:

$$\mathbb{I} \xrightarrow{m} \mathcal{O}(2) \quad \& \quad \bar{\mathbb{I}} \xrightarrow{e} \mathcal{O}(0)$$

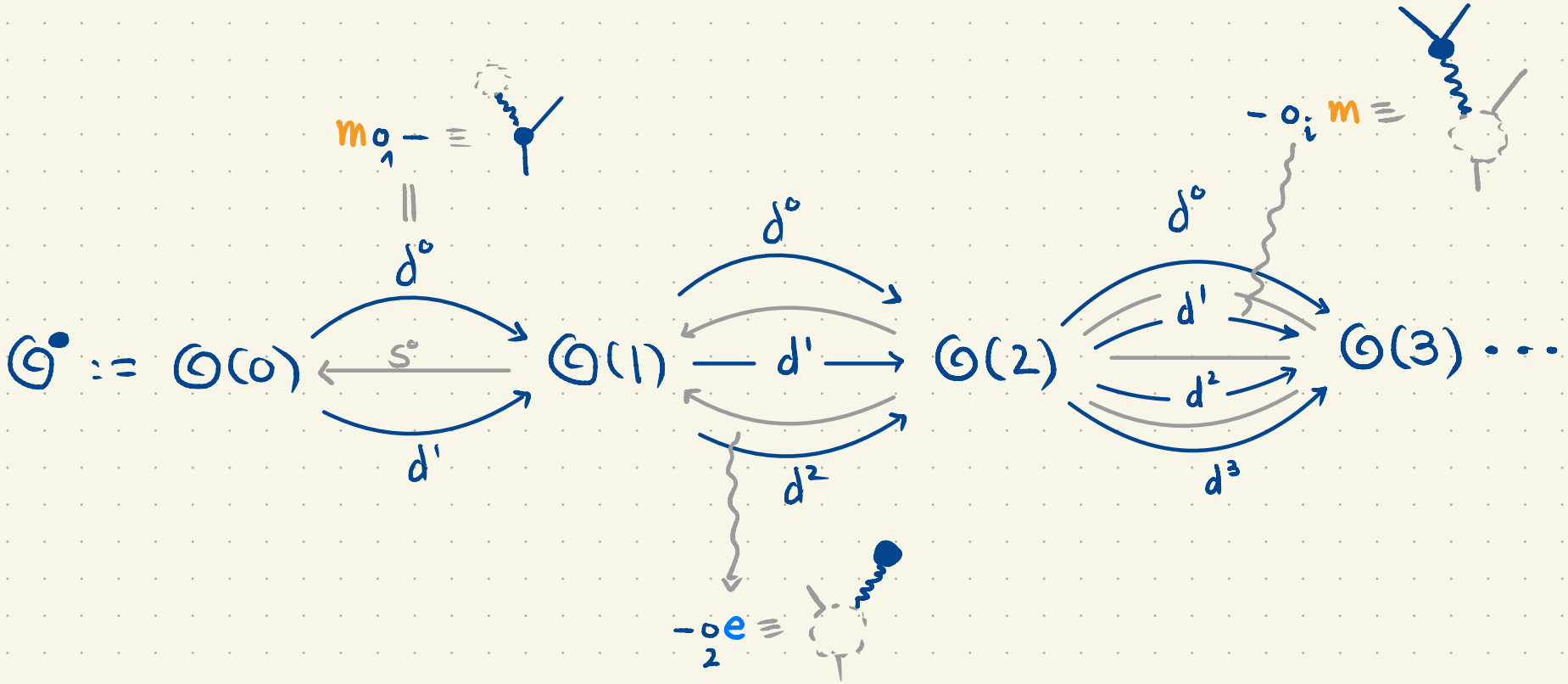
s.t.

$m \circ_1 m = m \circ_2 m$ (assoc.)

$m \circ e = id = m \circ_2 e$ (unital.)

Construction [Gerstenhaber-Voronov] There is a functorial rule

$$\left\{\begin{array}{c}\text{multiplicative}\\ \mathcal{V}\text{-operads}\end{array}\right\} \longrightarrow \mathcal{V}^{\Delta}, \quad (\mathcal{O},\eta) \longmapsto \mathcal{O}^{\bullet}$$



Rks: • There is an obvious dual construction:

$$\left\{\begin{array}{l}\text{comultiplicative}\\ \mathcal{V}\text{- co-operads}\end{array}\right\} \longrightarrow \mathcal{V}^{\Delta^{op}},\quad (\mathcal{C},\gamma)\longmapsto \mathcal{C}_\bullet$$

• If $\mathcal{V}=Ch_{\Bbbk}$ the totalization of $\mathcal{O}^\bullet$ coincides w/ the deformation complex of the map $\eta: As_+ \to \mathcal{O}$ $\rightsquigarrow$ $HH_*(\mathcal{O})$

(This would imply later a connection to $HH_*(\overline{\mathbb{P}}_d)$) — $(d-1)$-shifted Poisson opd.

• Also related: $\mathrm{Tot}\,\mathcal{O}^\bullet \simeq \Omega^2 \mathrm{Map}^h_{opd}(Ass_+, \mathcal{O})$

↑ if $\mathcal{O}(0)\simeq *\simeq \mathcal{O}(1)$

Given the unknot $\epsilon : \mathbb{R} \hookrightarrow \mathbb{R}^d$, we almost have a multipl. operad

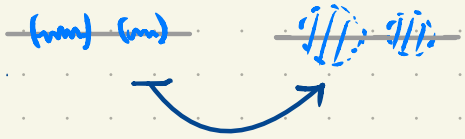
$t \longmapsto (t, 0 \dots 0)$

$$\mathbb{E}_1 \xrightarrow{\epsilon_\#} \mathbb{E}_d \qquad \mathbb{E}_1^{(0)} \xrightarrow{\epsilon_\#} \mathbb{E}_d^{us}$$

conn. comp. over $\mathrm{id} \in \Sigma_\bullet$.

$\simeq$

As_+



(or use FM_n if you like)

Sinha modified the FM-compactification so that

the compactification of $\mathrm{Conf}_{\mathbb{R}^1}(\bullet)$ is $Ass_+(\bullet) = \Sigma_\bullet$

$\Rightarrow \epsilon : \mathbb{R} \hookrightarrow \mathbb{R}^d \rightsquigarrow$ multiplicat. opd.

Consider

$$\mathcal{K}_d(r) := \text{closure of } \frac{Conf_{\mathbb{R}^d}(r)}{\mathbb{R}^d \rtimes \mathbb{R}} \xrightarrow{\text{relat. positions}} \prod_{1\le i<j\le r} S^{d-1}$$

translations and dilations

$$[x_1, \dots x_r] \longmapsto \left(\frac{x_i - x_j}{\|x_i - x_j\|}\right)_{i<j}$$

Recall

$$FM_d(r) := \text{closure of } \frac{Conf_{\mathbb{R}^d}(r)}{\mathbb{R}^d \rtimes \mathbb{R}} \xrightarrow{\text{relat. positions \& relat. dist.}} \prod_{1\le i<j\le r} S^{d-1} \times \prod_{1\le i<j<k\le r} [0,\infty]$$

Prop • $\mathcal{K}_d(r) \simeq Conf_{\mathbb{R}^d}(r)$

• $\mathcal{K}_d$ is an $\mathbb{E}_d$-operad in spaces and $\mathcal{K}_1 \cong Ass_+$

• $(\mathcal{K}_d, \epsilon_\pm : \mathcal{K}_1^{(0)} \to \mathcal{K}_d^{ns})$ is a multip. operad

Therefore, associated to $(\mathcal{K}_d, \epsilon_\#)$ we get:

$$\mathcal{K}_d^\bullet \in \mathrm{Spc}^{\Delta} \rightsquigarrow \operatorname*{holim}_{\Delta_{\le 0}} \mathcal{K}_d^\bullet|_{\le 0} \longleftarrow \operatorname*{holim}_{\Delta_{\le 1}} \mathcal{K}_d^\bullet|_{\le 1} \longleftarrow \underbrace{\operatorname*{holim}_{\Delta} \mathcal{K}_d^\bullet}_{\mathrm{holim}\, \mathcal{K}_d^\bullet}$$

coskelet. tower

Thm [Sinha] The GW-tower associated to $\overline{\mathrm{Emb}_c}(-, \mathbb{R}^d)$ (evaluat. on $\square'$) coincides with the tower above when $d \ge 3$:

$$\operatorname*{holim}_{\Delta_{\le k}} \mathcal{K}_d^\bullet|_{\le k} \simeq T_k \overline{\mathcal{K}nots}_d$$

In particular, if $d \ge 4$, $\operatorname*{holim}_{\Delta} \mathcal{K}_d^\bullet \simeq \overline{\mathcal{K}nots}_d$.

Corollary [Sinha, Turchin] For $d \geq 3$, the following spect. seqs coincide

$$\left[\begin{array}{c} E^2\text{-page of} \\ H_+^{\mathbb{Q}}\text{-BK-spect.seq} \\ \text{assoc. to } T.\overline{\mathcal{K}nots}_d \end{array}\right] \cong \left[\begin{array}{c} E^2\text{-page of} \\ H_+^{\mathbb{Q}}\text{-BK-spect.seq} \\ \text{assoc. to } \mathcal{K}_d^{\bullet} \end{array}\right] \cong \left[\begin{array}{c} E^1\text{-page of} \\ H_+^{\mathbb{Q}}\text{-Vassiliev s.s.} \\ \text{assoc. to } \overline{\mathcal{K}nots}_d \end{array}\right]$$

S11 Turchin

$$HH_*(\mathbb{P}_d)$$

Rk: Shifted Poisson is just: $\mathbb{P}_d \cong H_*^{\mathbb{Q}}(\mathbb{E}_d)$

Also, the collapse of BK-spect.seq. at $E^2 \Rightarrow$ $H_*^{\mathbb{Q}}(\overline{\mathcal{K}nots}_d) \overset{d \geq 4}{\cong} HH_*(\mathbb{P}_d)$

5. Relative $\mathbb{Q}$-formality implies collapse of BK-SS

We already know that

$$\mathcal{K}^\bullet \in \mathrm{Spc}^\Delta \ \mathbb{Q}\text{-formal} \Longrightarrow H^{\mathbb{Q}}_+ \text{-BK spect. seq. assoc. to } \mathcal{K}^\bullet \text{ collaps. at } E^2$$

Thus, if we prove the following implications

$$\underset{\text{is } \mathbb{Q}\text{-formal}}{\mathbb{E}_1 \xrightarrow{\epsilon_\#} \mathbb{E}_d} \overset{(1)}{\Longrightarrow} \underset{\text{is } \mathbb{Q}\text{-formal}}{As_+ \xrightarrow{\epsilon_\#} \mathcal{K}_d} \overset{(2)}{\Longrightarrow} \underset{\text{is } \mathbb{Q}\text{-formal}}{\mathcal{K}^\bullet_d \in \mathrm{Spc}^\Delta}$$

[Fresse-Willwacher] [Boavida-Horel]

we get:

Thm [Lambrechts-Turchin-Volic] For $d \geq 3$, the BK-spect.seq. associated to $T_\bullet \overline{\mathcal{K}nots}_d$ collapses at E^2 (hence, so does ... for $T_\bullet \mathcal{K}nots_d$)

Let us start w/ (2)

Lemma $\mathbb{Q}$-formality of $\epsilon_\sharp : As_+ \longrightarrow \mathcal{K}_d$ (i.e. as multipl. opd) $\Longrightarrow$ $\mathcal{K}_d^\bullet \in Spc^\Delta$ is $\mathbb{Q}$-formal

(Proof) The lhs means that there is a zigzag of weak equivalences in the category of maps of ns-operads (in $Ch(\mathbb{Q})$)

$$\begin{array}{ccccccccc} As_+ & = & As_+ & = & As_+ & \cdots & As_+ \\ \downarrow{\scriptstyle C_*(\epsilon_\sharp)} & & \downarrow & & \downarrow & & \downarrow{\scriptstyle H_*(\epsilon_\sharp)} \\ C_*^{\mathbb{Q}}(\mathcal{K}_d)^{ns} & \xleftarrow{\sim} & \mathcal{P}_1 & \xrightarrow{\sim} & \mathcal{P}_2 & \cdots & H_*^{\mathbb{Q}}(\mathcal{K}_d)^{ns} \end{array}$$

Gerstenhaber-Voronov's construction is functorial and preserves weqs

$$\Longrightarrow \quad C_*^{\mathbb{Q}}(\mathcal{K}_d^\bullet)^{ns} \xleftarrow{\sim} \mathcal{P}_1^\bullet \xrightarrow{\sim} \mathcal{P}_2^\bullet \cdots H_*^{\mathbb{Q}}(\mathcal{K}_d^\bullet)^{ns} \qquad \square$$

Finally, let us check (1)

Lemma [Songhafouo-Tsopmene] For $d \geq 3$, $(\mathcal{K}_d, \epsilon_\#)$ is $\mathbb{Q}$-formal (as a multip. opd.)

(Proof) As mentioned above, we use $\mathbb{Q}$-formality of $\mathbb{E}_1 \longrightarrow \mathbb{E}_d$

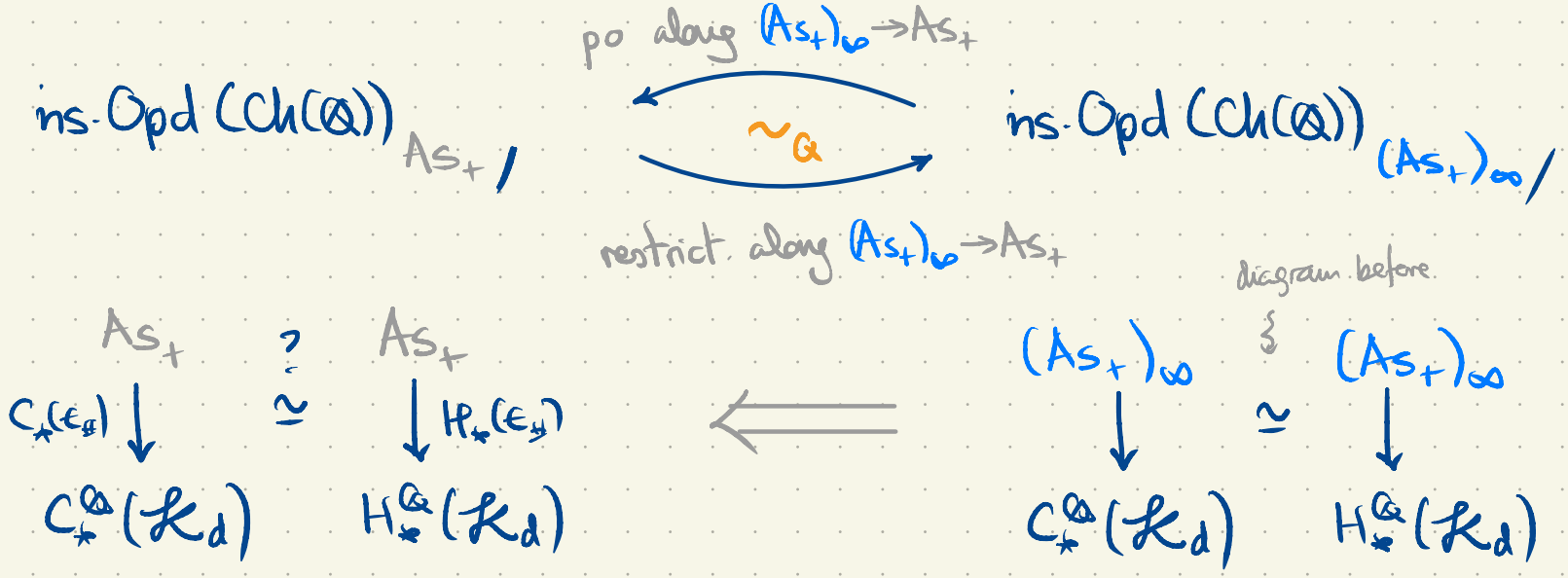
Actually, we have the following commutative diagram

$$\begin{array}{ccccccccc}
As_+ & \xleftarrow{\sim} & C_*^{\mathbb{Q}}(FM_1^{(0)}) & \xleftarrow{\sim} & (A_+)_\infty & \xrightarrow{\sim} & H_*^{\mathbb{Q}}(FM_1^{(0)}) & \xrightarrow{\simeq} & As_+ \\
\downarrow{\scriptstyle C_*(\epsilon_1)} & & \downarrow{\scriptstyle C_*(\epsilon_\#)} & & \downarrow & & \downarrow{\scriptstyle H_*(\epsilon_\#)} & & \downarrow{\scriptstyle H_*(\epsilon_\#)} \\
C_*^{\mathbb{Q}}(\mathcal{K}_d)^{ns} & \xleftarrow{\sim} & C_*^{\mathbb{Q}}(FM_d)^{ns} & \xleftarrow{\sim} & (\mathbb{E}_d)_\infty^{ns} & \xrightarrow{\sim} & H_*^{\mathbb{Q}}(FM_d)^{ns} & \xrightarrow{\simeq} & H_*^{\mathbb{Q}}(\mathcal{K}_d)^{ns}
\end{array}$$

By (relative) left-properness of ns-operads in $Ch(\mathbb{Q})$, we have a Quillen eq.

$$ns\text{-}Opd(Ch(\mathbb{Q}))_{As_+/} \underset{\text{restrict along } (As_+)_\infty \to As_+}{\overset{\text{po along } (As_+)_\infty \to As_+}{\sim_Q}} ns\text{-}Opd(Ch(\mathbb{Q}))_{(As_+)_\infty/}$$

By (relative) left-properness of ns-operads in $Ch(\mathbb{Q})$, we have a Quillen eq.



□

Again, (1) and (2) together, imply:

Thm [Lambrechts-Turchin-Volic] For $d \geq 3$, the BK-spect.seq. associated to $T_\bullet \overline{\mathcal{K}nots}_d$ collapses at E^2 (hence, so does ... for $T_\bullet \mathcal{K}nots_d$)

(in fact : $HH_*(P_d) \cong H_*^{\mathbb{Q}}(T_\infty \overline{\mathcal{K}nots}_d) \underset{d\geq 4}{\cong} H_*^{\mathbb{Q}}(\overline{\mathcal{K}nots}_d)$)

The $\pi_*^{\mathbb{Q}}$-BK.spect.seq. associated to $\mathcal{K}_d^\bullet$ also collapses at E^2. However, the proof of [Arone et al.] is more convoluted than the one before.

Idea: There is another $|X^\bullet| : \Delta \longrightarrow \mathcal{S}pc_*^{\geq 2}$ which is $\mathbb{Q}$-(co)formal and such that

$$\left[\begin{array}{c} H_*^{\mathbb{Q}}/\pi_*^{\mathbb{Q}}\text{-BK spect. seq.} \\ \text{assoc. to } |X^\bullet| \end{array}\right] \equiv \left[\begin{array}{c} H_*^{\mathbb{Q}}/\pi_*^{\mathbb{Q}}\text{-BK spect. seq.} \\ \text{assoc. to } \mathcal{K}_d^\bullet \end{array}\right]$$

Thm [Arone-Lambrechts-Turchin-Volic] If $d \geq 3$, $\pi_*^{\mathbb{Q}}$-BK spect.seq. assoc. to $\mathcal{K}_d^\bullet$ collapses at E^2 (equiv. $T.\overline{\mathcal{K}nots}_d$ & $T.\mathcal{K}nots_d$)

Rks: • [Boavida-Horel] deduced these collapse results, and even p-local / p-adic collapse in a range, thanks to GT-actions on $\overline{\mathbb{E}}_1 \xrightarrow{\epsilon_d} \overline{\mathbb{E}}_d$

(i.e. $GT_{\mathbb{Q}} \longrightarrow \pi_0 \mathrm{Aut}^h_{\mathrm{Opd}}(L_{\mathbb{Q}}\overline{\mathbb{E}}_n) \xrightarrow{\text{rest}} \pi_0 \mathrm{Aut}^h_{\mathrm{Spc}}(L_{\mathbb{Q}}\mathbb{E}_n(2)) \cong \mathbb{Q}^\times$)

• Combined w/ Goodwillie-Klein connectivity estimates, we get:

$$HH_*(\mathbb{P}_d) \underset{d\geq 4}{\simeq} H^{\mathbb{Q}}_*(\overline{\mathcal{K}nots_d}) \underset{\text{Künneth}}{\cong} H^{\mathbb{Q}}_*(\mathcal{K}nots_d) \otimes_{\mathbb{Q}} H^{\mathbb{Q}}_*(\Omega^2 S^{d-1})$$

$$\pi_i^{\mathbb{Q}}(\mathcal{K}nots_d) \cong \bigoplus_{t-s=i} E^2_{-s,t} \rightsquigarrow \overline{E}^2\text{-page of } \pi_*^{\mathbb{Q}}\text{-BK spect.seq. assoc. to } T_\bullet \mathcal{K}nots_d$$

6. Vassiliev invariants of knots

We have collapse of spect.seqs for $d \geq 3$, even though GW-tower does not converge for $d=3$

⚠ In fact, Goodwillie observed

$\pi_0(\mathcal{K}nots_3)$ is countable & $\pi_0(T_\infty \mathcal{K}nots_3)$ is uncountable

Hence, the tower does not converge even for the first group.

However, we can still extract interesting data from the collapse of the BK-spect.seq. at E^2 $\rightsquigarrow$ Vassiliev tower of fin. type knot-invariants.

Recall:

$$\pi_0(\mathrm{Emb}_c(\mathbb{R},\mathbb{R}^3)) \cong \pi_0(\mathrm{Emb}(S^1,S^3))$$

(1-pt cpt.)

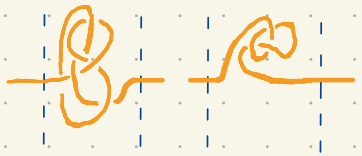
$\pi_0(\mathrm{Emb}_c(\mathbb{R},\mathbb{R}^3))$ = Knots_3

$\pi_0(\mathrm{Emb}(S^1,S^3))$ = classical isot. classes of knots

A modest community try to classify/distinguish knots. A general strategy is to construct knot invts. For instance,

Additive knot invariants $:= H^0(\pi_0(\mathrm{Knots}_3), A) \equiv$ monoid maps $\pi_0(\mathrm{Knots}_3) \longrightarrow A$ (abelian gp)

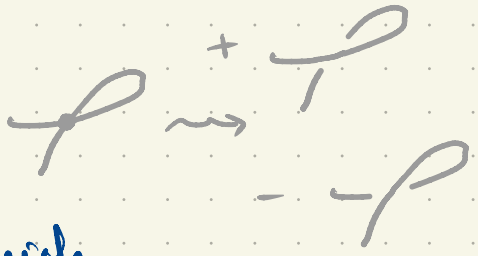
($\pi_0(\mathrm{Knots}_3)$: commutative monoid wrt)



$\uparrow$

Vassiliev/fin. type knot invts : those s.t. some extension to singular knots vanish

(finite # of double pts)



Filter fin.type knot invts by # of double pts $\rightsquigarrow$ deg$\leq n$ invts

(Def) $K \underset{n}{\sim} K'$ if $\psi(K) = \psi(K')$ $\forall\, \psi$ fin.type knot invt of deg$\leq n$

$\downsquigarrow$ Tower of fin.gen. abelian groups

$$\underset{\sim_1}{\pi_0(\mathcal{K}nots_3)} \leftarrow \underset{\sim_2}{\pi_0(\mathcal{K}nots_3)} \leftarrow \cdots \leftarrow \underset{\sim_k}{\pi_0(\mathcal{K}nots_3)} \leftarrow \cdots$$

On the other hand, we have the GW-tower for π_0 (also of f.g. ab.gps)

$$\pi_0(T_0\,\mathcal{K}nots_3) \leftarrow \pi_0(T_1\,\mathcal{K}nots_3) \leftarrow \pi_0(T_2\,\mathcal{K}nots_3) \leftarrow \cdots \leftarrow \pi_0(T_k\,\mathcal{K}nots_3) \leftarrow \cdots$$

$\shortparallel$ $\shortparallel$

0 0

These towers can be compared (up to a degree shift)

$$\begin{array}{ccccccccc}
\underset{\sim_1}{\pi_0(\mathcal{K}nots_3)} & \leftarrow & \underset{\sim_2}{\pi_0(\mathcal{K}nots_3)} & \leftarrow \cdots \leftarrow & \underset{\sim_{k-1}}{\pi_0(\mathcal{K}nots_3)} & \leftarrow \cdots \\
\downarrow \bar{e}_2 & & \downarrow \bar{e}_3 & & \downarrow \bar{e}_k & \\
\pi_0(T_2\mathcal{K}nots_3) & \leftarrow & \pi_0(T_3\mathcal{K}nots_3) & \leftarrow \cdots \leftarrow & \pi_0(T_k\mathcal{K}nots_3) & \leftarrow \cdots
\end{array}$$

Conjecture: Vassiliev tower $\cong$ $\pi_0(GW)$-tower for $\mathcal{K}nots_3$

Equivalently, $e_n : \pi_0(\mathcal{K}nots_3) \longrightarrow \pi_0(T_n\mathcal{K}nots_3)$

is the universal Vassiliev invariant of deg $\leq n-1$

Rk: Same question after $\otimes R$ w/ $R \in Ab$

Lemma [Kosanovic] Let $R \in Ab$ be torsion free. Then, the collapse of

$$\pi_*(-)\otimes R \equiv \pi_*^R \text{ - BK spect. seq. assoc. to } T_\bullet \mathcal{K}nots_3$$

at E^2 page along the diagonal implies that **Conjecture** is true after $\otimes R$

That is, $e_n : \pi_0(\mathcal{K}nots_3) \longrightarrow \pi_0(T_n \mathcal{K}nots_3)\otimes R$

is the universal Vassiliev invt of deg $\leq n-1$ over R

if $E^2_{-k,k}\otimes R \xrightarrow{\cong} E^\infty_{-k,k}\otimes R \;\; \forall k \leq n$

(Proof) Look at the map between towers,

$$\begin{array}{ccccc}
\dfrac{\pi_0(\mathcal{K}nots_3)}{\sim_{k-2}} & \longleftarrow & \dfrac{\pi_0(\mathcal{K}nots_3)}{\sim_{k-1}} & \hookleftarrow & \ker =: \Phi_{k-1} \\
\downarrow \bar{e}_{k-1} & & \downarrow \bar{e} & & \downarrow \\
\pi_0(T_{k-1}\mathcal{K}nots_3) & \longleftarrow & \pi_0(T_k\mathcal{K}nots_3) & \hookleftarrow & \ker =: E^{\infty}_{-k,k}
\end{array}$$

The map in gray fits into a commutative triangle

$$E^2_{-k,k} \xrightarrow{\text{canon.}} E^{\infty}_{-k,k}, \qquad E^2_{-k,k} \twoheadrightarrow \Phi_{k-1} \hookrightarrow E^{\infty}_{-k,k}$$

[Conant - Teichner]

$\Rightarrow$ Apply $\otimes R$ and use 5-lemma to get:

$$\Phi_{k-1} \otimes R \xrightarrow{\cong} E^{\infty}_{-k,k} \otimes R$$

if

$$E^2_{-k,k} \otimes R \xrightarrow{\text{can.}} E^{\infty}_{-k,k} \otimes R$$

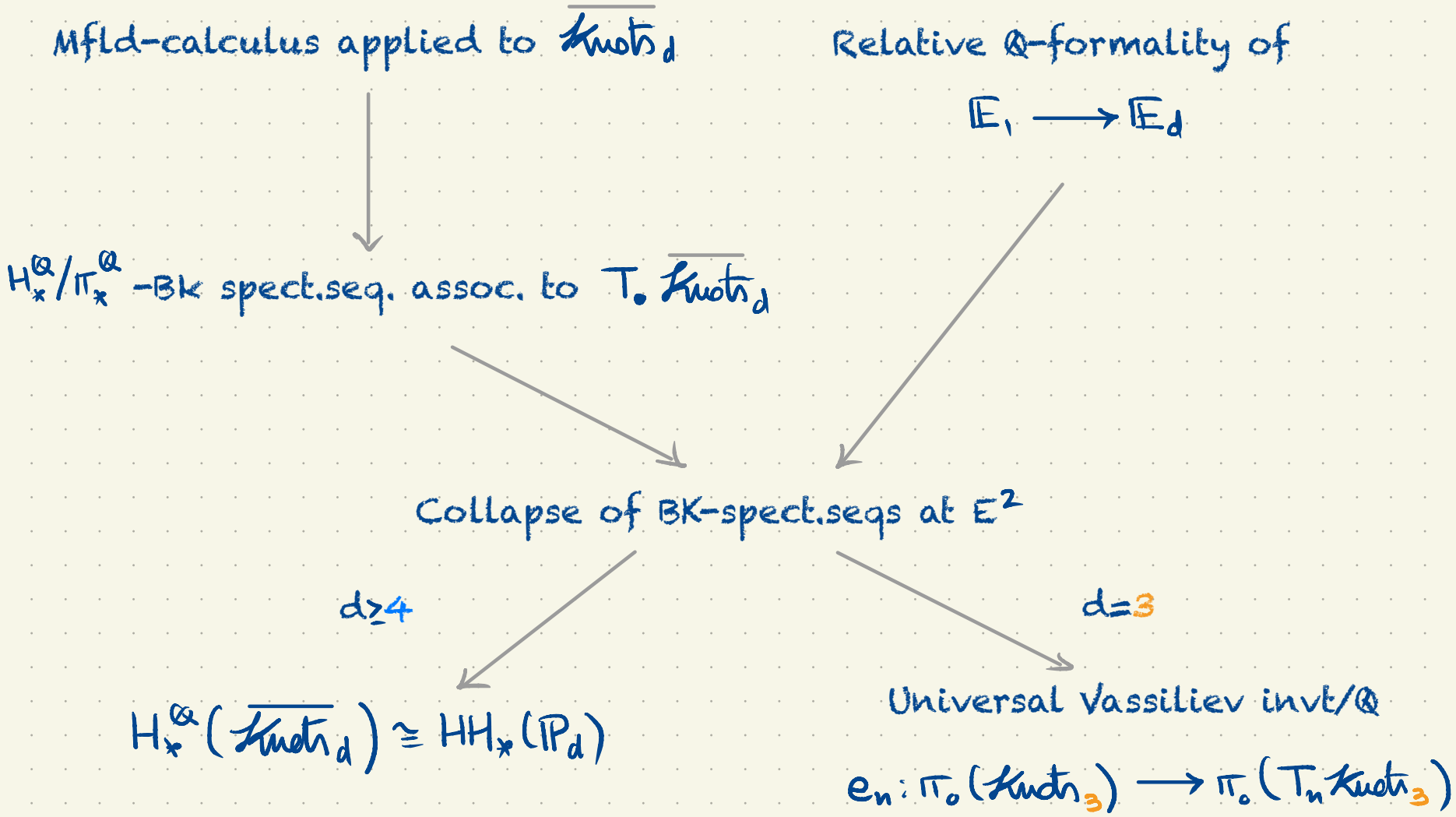
Since this holds $\forall$ $k \leq n$, we are done

Corollary $\pi_0(\mathcal{K}nots_3) \xrightarrow{e_n} \pi_0(\mathcal{K}nots_3)\otimes\mathbb{Q}$ is the univ. Vassiliev invt of deg $\leq n-1$ over $\mathbb{Q}$

Rk: • [Boavida-Horel] showed that this also holds for $R = \mathbb{Z}_{(p)}$ & $\mathbb{Z}_p$
($\mathbb{Z}_{(p)}$: p-local, $\mathbb{Z}_p$: p-adic)
in a range that increases w/ p

• It is also known for $R=\mathbb{Z}$ when $n\leq 7$ and by [Kosanovic], we know surjectivity of $\pi_0(\mathcal{K}nots_3)/_{\sim_{n-1}} \longrightarrow \pi_0(T_n\mathcal{K}nots_3)$ $\forall n$

Summary:


Mfld-calculus applied to $\overline{\mathbb{K}nots}_d$
Relative $\mathbb{Q}$-formality of
$\mathbb{E}_1 \longrightarrow \mathbb{E}_d$
$H_*^{\mathbb{Q}}/\pi_*^{\mathbb{Q}}$-Bk spect.seq. assoc. to $T_\bullet \overline{\mathbb{K}nots}_d$
Collapse of BK-spect.seqs at E^2
$d \geq 4$
$d = 3$
$H_*^{\mathbb{Q}}(\overline{\mathbb{K}nots}_d) \cong HH_*(\mathbb{P}_d)$
Universal Vassiliev invt/$\mathbb{Q}$
$e_n : \pi_0(\mathbb{K}nots_3) \longrightarrow \pi_0(T_n \mathbb{K}nots_3)$

That was all, folks